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Multilayer electronic component having external electrode including alloy layer between base electrode layer and plating layer

Abstract

A multilayer electronic component includes a body including a dielectric layer and an internal electrode; and an external electrode including a base electrode layer disposed on the body and including a first metal element, and an alloy layer and a plating layer, sequentially disposed on the base electrode layer. The alloy layer includes a first alloy layer disposed to contact the base electrode layer and including an alloy of the first metal element and Sn, and a second alloy layer disposed to contact the first alloy layer and including an alloy of Ni and Sn, and the plating layer includes an Ni plating layer disposed to contact the second alloy layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (S)

(1) This application claims benefit of priority to Korean Patent Application No. 10-2023-0033987 filed on Mar. 15, 2023 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a multilayer electronic component.

(3) A multilayer ceramic capacitor (MLCC), a multilayer electronic component, may be a chip-type condenser mounted on the printed circuit boards of any of various types of electronic products, such as an image display device, including a liquid crystal display (LCD), a plasma display panel (PDP), or the like, a computer, a smartphone, or a mobile phone, serving to charge or discharge electricity therein or therefrom.

(4) The multilayer ceramic capacitor has a small size, implements high capacitance, and is easily mounted, and may thus be used as a component of various electronic devices.

(5) In general, an external electrode of a multilayer ceramic capacitor may include a base electrode

layer, an Ni plating layer disposed on the base electrode layer, and an Sn plating layer disposed on the Ni plating layer.

(6) When the multilayer ceramic capacitor is mounted on a substrate, tensile stress or physical stress caused by heat generated during a reflow process may be applied to the multilayer ceramic capacitor. In this case, an interface between the base electrode layer and the Ni plating layer, a portion having weak adhesion, may be peeled off.

(7) In order to solve this problem, Patent Document 1 has tried to improve adhesion between the base electrode layer and the Ni plating layer by forming an Ni—Sn alloy plating layer between the base electrode layer and the Ni plating layer.

(8) However, since it is difficult to improve adhesion between the base electrode layer and the Ni—Sn alloy plating layer only by forming the Ni—Sn alloy plating layer, a new method is required.

(Patent Document 1) JP 2021-100019 A

SUMMARY

(9) An aspect of the present disclosure is to provide a multilayer electronic component having improved adhesion between a base electrode layer and an Ni plating layer.

(10) An object of the present disclosure is not limited to the above, and will be more easily understood in the process of describing specific embodiments of the present disclosure.

(11) According to an aspect of the present disclosure, a multilayer electronic component includes a body including a dielectric layer and an internal electrode; and an external electrode including a base electrode layer disposed on the body and including a first metal element, and an alloy layer and a plating layer, sequentially disposed on the base electrode layer. The alloy layer includes a first alloy layer disposed to contact the base electrode layer and including an alloy of the first metal element and Sn, and a second alloy layer disposed to contact the first alloy layer and including an alloy of Ni and Sn, and the plating layer includes an Ni plating layer disposed to contact the second alloy layer.

(12) According to an aspect of the present disclosure, a multilayer electronic component includes a body including a dielectric layer and an internal electrode; and an external electrode including a base electrode layer disposed on the body and including a first metal element, and an alloy layer and a Ni plating layer, sequentially disposed on the base electrode layer. The alloy layer includes an alloy of the first metal element, Sn, and Ti. An average thickness of the alloy layer is 0.2 μm or less.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings.

(2) FIG. 1 is a perspective view schematically illustrating a multilayer electronic component according to an embodiment of the present disclosure.

(3) FIG. 2 is a cross-sectional view schematically illustrating the multilayer electronic component of FIG. 1, taken along line I-I'.

(4) FIG. 3 is a cross-sectional view schematically illustrating the multilayer electronic component of FIG. 1, taken along line II-II'.

(5) FIG. 4 is an exploded perspective view schematically illustrating a body.

(6) FIG. 5 is an enlarged view of portion K of FIG. 2.

(7) FIG. 6 is an image of a cross-section of an external electrode of a multilayer electronic component according to an embodiment of the present disclosure, scanned with a scanning electron microscope.

(8) FIG. 7 is a graph obtained by performing a line profile along line L1 of FIG. 6 using SEM-EDS.

(9) FIG. 8 is an image of a cross-section of an external electrode of an inventive example, scanned with a scanning electron microscope.

(10) FIG. 9 is an image of a cross-section of an external electrode of a comparative example, scanned with a scanning electron microscope.

DETAILED DESCRIPTION

(11) Hereinafter, embodiments of the present disclosure will be described with reference to specific example embodiments and the attached drawings. The embodiments of the present disclosure may, however, be exemplified in many different forms and should not be construed as being limited to the specific embodiments set forth herein. The example embodiments disclosed herein are provided for those skilled in the art to better explain the present disclosure. In the drawings, the shapes and dimensions of elements may be exaggerated for clarity, and the same reference numerals will be used throughout to designate the same or like elements.

(12) In addition, in order to clearly describe the present disclosure in the drawings, the contents unrelated to the description are omitted, and since sizes and thicknesses of each component illustrated in the drawings are arbitrarily illustrated for convenience of description, the present disclosure is not limited thereto. In addition, components with the same function within the same range of ideas are described using the same reference numerals. Throughout the specification, when a certain portion “includes” or “comprises” a certain component, this indicates that other components are not excluded and may be further included unless otherwise noted.

(13) In the drawings, a first direction may be defined as a thickness T direction, a second direction may be defined as a length L direction, and a third direction may be defined as a width W direction.

(14) Multilayer Electronic Component

(15) FIG. 1 is a perspective view schematically illustrating a multilayer electronic component according to an embodiment of the present disclosure.

(16) FIG. 2 is a cross-sectional view schematically illustrating the multilayer electronic component of FIG. 1, taken along line I-I'.

(17) FIG. 3 is a cross-sectional view schematically illustrating the multilayer electronic component of FIG. 1, taken along line II-II'.

(18) FIG. 4 is an exploded perspective view schematically illustrating a body.

(19) FIG. 5 is an enlarged view of portion K of FIG. 2.

(20) Hereinafter, a multilayer electronic component **100** according to an embodiment of the present disclosure will be described in detail with reference to FIGS. 1 to 5. In addition, multilayer ceramic capacitor (hereinafter referred to as ‘MLCC’) will be described as an example of the multilayer electronic component, but the present disclosure is not limited thereto, and may be applied to various multilayer electronic components having an external electrode, such as an inductor, a piezoelectric element, a varistor, a thermistor, or the like.

(21) A multilayer electronic component **100** according to an embodiment of the present disclosure may include a body **110** including a dielectric layer **111** and an internal electrode (**121** and **122**); an external electrode (**131** and **132**) including a base electrode layer (**131a** and **132a**) disposed on the body and including a first metal element, an alloy layer (**131b** and **132b**) disposed on the base electrode layer, and a plating layer (**131c** and **132c**) disposed on the alloy layer (**131b** and **132b**). The alloy layer (**131b**) may include a first alloy layer (**131b1**) disposed to contact the base electrode layer (**131a**) and including an alloy of the first metal element and Sn, and a second alloy layer (**131b2**) disposed to contact the first alloy layer and including an alloy of Ni and Sn. The plating layer **131c** may include an Ni plating layer (**131c1**) disposed to contact the second alloy layer (**131b2**). The alloy layer (**132b**) may be configured similar to the alloy layer (**131b**) to have a first alloy layer disposed to contact the base electrode layer (**132a**) and including an alloy of the first metal element and Sn, and a second alloy layer disposed to contact the first alloy layer and

including an alloy of Ni and Sn. The plating layer (**132c**) may be configured similar to the plating layer (**131c**) and may include an Ni plating layer disposed to contact the second alloy layer of the alloy layer (**132b**). The detailed structure of the alloy layer (**132b**) and the plating layer (**132c**), and the layers/materials constituting the alloy layer (**132b**) and the plating layer (**132c**) may be respectively referred to the alloy layer (**131b**) and the plating layer (**131c**), and thus, the redundant descriptions will be omitted.

(22) As described above, when the multilayer ceramic capacitor, which is a multilayer electronic component, is mounted on a substrate, tensile stress or physical stress caused by heat generated during a reflow process may be applied to the multilayer ceramic capacitor. In this case, an interface between the base electrode layer and the Ni plating layer, which is a portion having weak adhesion, may be peeled off.

(23) In order to solve this problem, a method of improving adhesion between a base electrode layer and an Ni plating layer by forming an Ni—Sn alloy plating layer between the base electrode layer and the Ni plating layer has been proposed. In this case, adhesion may be improved, as compared to a case in which the Ni—Sn alloy plating layer is not provided, but there may be a limit in improving adhesion between the Ni—Sn alloy plating layer and the base electrode layer. In addition, since a composition of the Ni—Sn alloy plating layer may be clearly changed on an interface with the base electrode layer and an interface with the Ni plating layer, an effect of improving the adhesion may be limited. On an interface between heterogeneous metals, as compositions thereof are similar to each other, adhesion therebetween may increase. When the Ni—Sn alloy plating layer is formed on the base electrode layer, since compositions thereof may be rapidly changed on an interface between the base electrode layer and the Ni—Sn alloy plating layer, adhesion therebetween may decrease.

(24) Therefore, since the present disclosure includes the first alloy layer (**131b1**) disposed to contact the base electrode layer (**131a**) and including the alloy of the first metal element and Sn, and the second alloy layer (**131b2**) disposed to contact the first alloy layer (**131b1**) and including the alloy of Ni and Sn, adhesion between the alloy layer (**131b1**) and the base electrode layer (**131a**) and adhesion between the alloy layer (**131b1** and **131b2**) and the Ni plating layer (**131c1**) may be improved at the same time, to improve adhesion between the base electrode layer (**131a**) and the Ni plating layer (**131c1**). For example, in the present disclosure, it is intended to improve adhesion between heterogeneous layers by making compositions of each other as similar as possible on an interface between the heterogeneous layers.

(25) Hereinafter, each component included in a multilayer electronic component **100** according to an embodiment of the present disclosure will be described.

(26) In the body **110**, the dielectric layer **111** and the internal electrode (**121** and **122**) may be alternately stacked.

(27) Although the specific shape of the body **110** is not particularly limited, as illustrated, the body **110** may have a hexahedral shape or a shape similar thereto. Due to shrinkage of ceramic powder particles included in the body **110** during a sintering process, the body **110** may not have a perfectly straight hexahedral shape, but may have a substantially hexahedral shape.

(28) The body **110** may include first and second surfaces **1** and **2** opposing each other in the first direction, third and fourth surfaces **3** and **4** connected to the first and second surfaces **1** and **2** and opposing each other in the second direction, and fifth and sixth surfaces **5** and **6** connected to the first and second surfaces **1** and **2**, connected to the third and fourth surfaces **3** and **4**, and opposing each other in the third direction.

(29) As a margin region on which the internal electrode (**121** and **122**) is not disposed to overlap the dielectric layer **111**, a step difference may be generated due to a thickness of the internal electrode (**121** and **122**), such that a corner connecting the first surface and the third to fifth surfaces and/or a corner connecting the second surface and the third to fifth surfaces have a shape contracted toward a center of the body **110** in the first direction, when viewed from the first surface

or the second surface. Alternatively, due to contract behavior of the body during a sintering process, a corner connecting the first surface **1** and the third to sixth surfaces **3**, **4**, **5**, and **6** and/or a corner connecting the second surface **2** and the third to sixth surfaces **3**, **4**, **5**, and **6** may have a shape contracted toward the center of the body **110** in the first direction, when viewed from the first or second surface. Alternatively, to prevent a chipping defect or the like, edges connecting each surface of the body **110** may be rounded by performing a separate process, such that a corner connecting the first surface and the third to sixth surfaces and/or a corner connecting the second surface and the third to sixth surfaces may have a round shape.

(30) To suppress the step difference caused by the internal electrode (**121** and **122**), after stacking, the internal electrodes may be cut to be exposed from the fifth and sixth surfaces **5** and **6** of the body, and, when a margin portion (**114** and **115**) is formed by stacking a single dielectric layer or two or more dielectric layers on both side surfaces of a capacitance formation portion **Ac** in the third direction (width direction), a portion connecting the first surface **1** and the fifth and sixth surfaces **5** and **6** and a portion connecting the second surface **2** and the fifth and sixth surfaces **5** and **6** may not have a contracted shape.

(31) A plurality of dielectric layers **111** forming the body **110** may be in a sintered state, and a boundary between adjacent dielectric layers **111** may be integrated to such an extent that it is difficult to identify the same without using a scanning electron microscope (SEM). The number of stacked dielectric layers is not particularly limited, and may be determined in consideration of a size of the multilayer electronic component. For example, the body may be formed by stacking 400 or more dielectric layers.

(32) The dielectric layer **111** may be prepared by forming a ceramic slurry containing ceramic powder particles, an organic solvent, and a binder, coating and drying the slurry on a carrier film to prepare a ceramic green sheet, and then sintering the ceramic green sheet. The ceramic powder particles are not particularly limited as long as sufficient capacitance is obtained, but, for example, barium titanate-based ($\text{BaTiO}_{3.3}$)-based powder particles may be used as the ceramic powder particles. For more specific examples, the ceramic powder particles may be any one of $\text{BaTiO}_{3.3}$, $(\text{Ba}_{1-x}\text{Ca}_x)\text{TiO}_{3.3}$ ($0 < x < 1$), $\text{Ba}(\text{Ti}_{1-y}\text{Ca}_y)\text{O}_{3.3}$ ($0 < y < 1$), $(\text{Ba}_{1-x}\text{Ca}_x)(\text{Ti}_{1-y}\text{Zr}_y)\text{O}_{3.3}$ ($0 < x < 1$, $0 < y < 1$), or $\text{Ba}(\text{Ti}_{1-y}\text{Zr}_y)\text{O}_{3.3}$ ($0 < y < 1$).

(33) An average thickness t_d of the dielectric layer **111** is not particularly limited, but may be, for example, 10 μm or less. In addition, the average thickness t_d of the dielectric layer **111** may be arbitrarily set according to desired characteristics or uses. In addition, to more easily achieve miniaturization and high capacitance of the multilayer electronic component, the average thickness t_d of the dielectric layer **111** may be 0.4 μm or less.

(34) In this case, the average thickness t_d of the dielectric layer **111** means a size of the dielectric layer **111** disposed between the internal electrodes **121** and **122** in the first direction. The average thickness of the dielectric layer **111** may be measured by scanning an image of a cross-section of the body **110** in the first and second directions with a scanning electron microscope (SEM) at a magnification of 10,000. More specifically, an average value thereof may be determined by measuring a thickness of one (1) dielectric layer at thirty (30) equally spaced points in the second direction in the scanned image. The thirty (30) equally spaced points may be designated in the capacitance formation portion **Ac**, which will be described later. In addition, when such an average value is determined by extensively using measurements of average values to ten (10) dielectric layers, the average thickness of the dielectric layer **111** may be more generalized.

(35) The body **110** may include a capacitance formation portion **Ac** disposed in the body **110** and including a first internal electrode **121** and a second internal electrode **122** disposed to face each other with the dielectric layer **111** interposed therebetween to form capacitance, and cover portions **112** and **113** formed above and below the capacitance formation portion **Ac** in the first direction.

(36) In addition, the capacitance formation portion **Ac** may be a portion contributing to capacitance

formation of the capacitor, and may be formed by repeatedly stacking the plurality of first and second internal electrodes **121** and **122** with the dielectric layer **111** interposed therebetween.

(37) The cover portions **112** and **113** may include an upper cover portion **112** disposed above the capacitance formation portion **Ac** in the first direction, and a lower cover portion **113** disposed below the capacitance formation portion **Ac** in the first direction.

(38) The upper cover portion **112** and the lower cover portion **113** may be formed by stacking a single dielectric layer or two or more dielectric layers on the upper and lower surfaces of the capacitance formation portion **Ac** in the thickness direction, respectively, and may basically play a role in preventing damage to the internal electrodes due to physical or chemical stress.

(39) The upper cover portion **112** and the lower cover portion **113** may not include internal electrodes, and may include the same material as the dielectric layer **111**.

(40) For example, the upper cover portion **112** and the lower cover portion **113** may include a ceramic material, for example, a barium titanate (BaTiO₃)-based ceramic material.

(41) A thickness of each of the cover portions **112** and **113** does not need to be particularly limited. A thickness t_c of each of the cover portions **112** and **113** may be 15 μm or less to more easily achieve miniaturization and high capacitance of the multilayer electronic component.

(42) An average thickness t_c of each of the cover portions **112** and **113** may mean a size in the first direction, and may be an average value of sizes in the first direction, measured at five (5) equally spaced points above or below the capacitance formation portion **Ac**.

(43) In addition, margin portions **114** and **115** may be disposed on side surfaces of the capacitance formation portion **Ac**.

(44) The marginal portions **114** and **115** may include a first marginal portion **114** disposed on the fifth surface **5** of the body **110**, and a second marginal portion **115** disposed on the sixth surface **6** of the body **110**. For example, the margin portions **114** and **115** may be disposed on both end surfaces of the body **110** in the width direction.

(45) As illustrated in FIG. 3, the margin portions **114** and **115** may refer to regions between both ends of the first and second internal electrodes **121** and **122** and a boundary surface of the body **110**, in a cross-section of the body **110** cut in the width-thickness (W-T) directions.

(46) The margin portions **114** and **115** may basically serve to prevent damage to the internal electrodes due to physical or chemical stress, respectively.

(47) The margin portions **114** and **115** may be prepared by forming internal electrodes by applying a conductive paste on a ceramic green sheet, except for regions in which the margin portions are formed.

(48) In addition, to suppress a step difference caused by the internal electrodes **121** and **122**, the margin portions **114** and **115** may be cut to expose the fifth and sixth surfaces **5** and **6** of the body after stacking, respectively, and may be then formed by stacking a single dielectric layer or two or more dielectric layers in the third direction (width direction) on both end surfaces of the capacitance formation portion **Ac**.

(49) Widths of the margin portions **114** and **115** are not particularly limited. An average width of each of the margin portions **114** and **115** may be 15 μm or less to more easily achieve miniaturization and high capacitance of the multilayer electronic component.

(50) The average width of each of the margin portions **114** and **115** may mean an average size **MW1** of each of the margin portions **114** and **115** in the third direction, in a region in which the internal electrode is spaced apart from the fifth surface, and an average size **MW2** of each of the margin portions **114** and **115** in the third direction, in a region in which the internal electrode is spaced apart from the sixth surface, respectively, and may be an average value of sizes of each of the margin portions **114** and **115** in the third direction, measured at five (5) equally spaced points on a side surface of the capacitance formation portion **Ac**.

(51) Therefore, in an embodiment, the average sizes **MW1** and **MW2** in the regions in which the internal electrodes **121** and **122** are spaced apart from the fifth and sixth surfaces in the third

direction may be 15 μm or less, respectively.

(52) The internal electrode (**121** and **122**) may include first and second internal electrodes **121** and **122**. The first and second internal electrodes **121** and **122** may be alternately disposed to face each other with the dielectric layer **111**, constituting the body **110**, interposed therebetween, and may be exposed from the third and fourth surfaces **3** and **4** of the body **110**, respectively.

(53) The first internal electrode **121** may be spaced apart from the fourth surface **4** and be exposed from (or extend from or be in contact with) the third surface **3**, and the second internal electrode **122** may be spaced apart from the third surface **3** and exposed from (or extend from or be in contact the fourth surface **4**. A first external electrode **131** may be disposed on the third surface **3** of the body and connected to the first internal electrode **121**, and a second external electrode **132** may be disposed on the fourth surface **4** of the body and connected to the second internal electrode **122**.

(54) For example, the first internal electrode **121** may be connected to the first external electrode **131** without being connected to the second external electrode **132**, and the second internal electrode **122** may be connected to the second external electrode **132** without being connected to the first external electrode **131**. Therefore, the first internal electrode **121** may be formed to be spaced apart from the fourth surface **4** by a predetermined distance, and the second internal electrode **122** may be formed to be spaced apart from the third surface **3** by a predetermined distance. Also, the first and second internal electrodes **121** and **122** may be spaced apart from the fifth and sixth surfaces of the body **110**.

(55) A conductive metal included in the internal electrode (**121** and **122**) may be at least one of Ni, Cu, Pd, Ag, Au, Pt, In, Sn, Al, Ti, or alloys thereof, but the present disclosure is not limited thereto.

(56) A method of forming the internal electrode (**121** and **122**) is not particularly limited. For example, the internal electrode (**121** and **122**) may be formed by applying a conductive paste for internal electrodes containing a conductive metal on a ceramic green sheet and sintering the same. As a method of applying the conductive paste for internal electrodes, a screen printing method, a gravure printing method, or the like may be used, but the present disclosure is not limited thereto.

(57) As another example, the internal electrode (**121** and **122**) may be formed using a sputtering method, a vacuum deposition method, and/or a chemical vapor deposition method.

(58) An average thickness t_e of the internal electrode does not need to be particularly limited. In this case, a thickness of the internal electrode (**121** and **122**) may mean a size of the internal electrode (**121** and **122**) in the first direction. For example, the average thickness t_e of the internal electrode (**121** and **122**) may be 0.4 μm or less to more easily achieve miniaturization and high capacitance of the multilayer electronic component.

(59) In this case, the average thickness t_e of the internal electrode may be measured by scanning an image of a cross-section of the body **110** in the first and second directions with a scanning electron microscope (SEM) at a magnification of 10,000. More specifically, an average value thereof may be determined by measuring a thickness of one (1) internal electrode at thirty (30) equally spaced points in the second direction in the scanned image. The thirty (30) equally spaced points may be designated in the capacitance formation portion A_c . In addition, when such an average value is determined by extensively using measurements of average values to ten (10) internal electrodes, the average thickness of the internal electrode may be further generalized.

(60) The external electrodes **131** and **132** may be disposed on the third and fourth surfaces **3** and **4** of the body **110**.

(61) The external electrodes **131** and **132** may be disposed on the third and fourth surfaces **3** and **4** of the body **110**, respectively, and may include first and second external electrodes **131** and **132** respectively connected to the first and second internal electrodes **121** and **122**.

(62) In the present embodiment, a structure in which the multilayer electronic component **100** has two external electrodes **131** and **132** will be described, but the number, shapes, and the like of the external electrodes **131** and **132** may depend on shapes of the internal electrodes **121** and **122**, and may be changed for other purposes.

(63) The external electrode (**131** and **132**) may include the base electrode layer (**131a** and **132a**) disposed on the body **110** and including the first metal element, the alloy layer (**131b** and **132b**) disposed on the base electrode layer, and the plating layer (**131c** and **132c**) disposed on the alloy layer.

(64) The alloy layer (**131b**) may include the first alloy layer (**131b1**) disposed to contact the base electrode layer (**131a**) and including the alloy of the first metal element and Sn, and the second alloy layer (**131b2**) disposed to contact the first alloy layer (**131b1**) and including the alloy of Ni and Sn, and the plating layer (**131c**) may include an Ni plating layer (**131c1**) disposed to contact the second alloy layer (**131b2**). For example, the external electrode (**131**) may have a structure in which the base electrode layer (**131a**), the first alloy layer (**131b1**), the second alloy layer (**131b2**), and the Ni plating layer (**131c1**) are sequentially arranged.

(65) The first alloy layer (**131b1**) may include the alloy of the first metal element and Sn to improve adhesion to the base electrode layer (**131a**) including the first metal element, and the second alloy layer (**131b2**) may include the alloy of Ni and Sn to improve adhesion to the Ni plating layer (**131c1**).

(66) In addition, the first alloy layer (**131b1**) and the second alloy layer (**131b2**) may include Sn in common to improve adhesion between the first alloy layer (**131b1**) and the second alloy layer (**131b2**).

(67) In an embodiment, a first metal element amount of the first alloy layer (**131b1**) may be highest on an interface with the base electrode layer (**131a**), and may be lowest on an interface with the second alloy layer (**131b2**). Therefore, adhesion on an interface between the first alloy layer (**131b1**) and the base electrode layer (**131a**) may be further improved, and the adhesion may be further improved by making a composition of the first alloy layer (**131b1**) and a composition of the second alloy layer (**131b2**) as similar as possible on an interface between the first alloy layer (**131b1**) and the second alloy layer (**131b2**).

(68) More preferably, the first metal element amount of the first alloy layer (**131b1**) may decrease from the interface with the base electrode layer (**131a**) toward the interface with the second alloy layer (**131b2**). As the first metal element amount of the first alloy layer (**131b1**) continuously changes, adhesion between the base electrode layer (**131a**) and the second alloy layer (**131b2**) may be further improved.

(69) In an embodiment, an Ni amount of the second alloy layer (**131b2**) may be highest on an interface with the Ni plating layer (**131c1**), and may be lowest on an interface with the first alloy layer (**131b1**). Therefore, adhesion on an interface between the second alloy layer (**131b2**) and the first alloy layer (**131b1**) may be further improved, and the adhesion may be further improved by making a composition of the second alloy layer (**131b2**) and a composition of the Ni plating layer (**131c1**) as similar as possible on an interface between the second alloy layer (**131b2**) and the Ni plating layer (**131c1**).

(70) More preferably, the Ni amount of the second alloy layer (**131b2**) may decrease from the interface with the Ni plating layer (**131c1**) toward the interface with the first alloy layer (**131b1**). As the Ni amount of the second alloy layer (**131b2**) continuously changes, adhesion between the first alloy layer (**131b1**) and the Ni plating layer (**131c1**) may be further improved.

(71) In an embodiment, a first metal element amount of the alloy layer (**131b**) may be highest on an interface with the base electrode layer (**131a**), and may be lowest on an interface with the Ni plating layer (**131c1**). An Ni amount of the alloy layer (**131b**) may be lowest on an interface with the base electrode layer (**131a**), and may be highest on an interface with the Ni plating layer (**131c1**). Therefore, the adhesion may be further improved by making a composition of the alloy layer (**131b** and **132b**) and a composition of the base electrode layer (**131a** and **132a**) as similar as possible on an interface between the alloy layer (**131b** and **132b**) and the base electrode layer (**131a** and **132a**), and by making a composition of the alloy a layer (**131b** and **132b**) and a composition of the Ni plating layer (e.g., **131c1**) as similar as possible on an interface between the alloy layer

(**131b** and **132b**) and the Ni plating layer (e.g., **131c1**).

(72) More preferably, the first metal element amount of the alloy layer (**131b** and **132b**) may decrease gradually from the interface with the base electrode layer (**131a** and **132a**) toward the interface with the Ni plating layer (e.g., **131c1**), and the Ni amount of the alloy layer (**131b** and **132b**) may increase from the interface with the base electrode layer (**131a** and **132a**) toward the interface with the Ni plating layer (e.g., **131c1**). As the first metal element amount and the Ni amount of the alloy layer (**131b** and **132b**) continuously change, adhesion to the base electrode layer (**131a** and **132a**) and the Ni plating layer (e.g., **131c1**) may be further improved.

(73) In an embodiment, the alloy layer (**131b**) may have the highest Sn amount on an interface between the first alloy layer (**131b1**) and the second alloy layer (**131b2**). Therefore, the adhesion may be further improved by making a composition of the first alloy layer (**131b1**) and a composition of the second alloy layer (**131b2**) as similar as possible on an interface between the first alloy layer (**131b1**) and the second alloy layer (**131b2**).

(74) In an embodiment, an Sn amount of the alloy layer (**131b**) may increase from an interface with the base electrode layer (**131a**) to an interface between the first alloy layer (**131b1**), and may decrease from an interface between the first alloy layer (**131b1**) and the second alloy layer (**131b2**) toward an interface with the Ni plating layer (**131c1**). As the Sn amount of the alloy layer (**131b**) continuously changes, adhesion between the base electrode layer (**131a**) and the Ni plating layer (**131c1**) may be further improved.

(75) A method of measuring an element amount of the first alloy layer (**131b1**) and an element amount of the second alloy layer (**131b2**) is not particularly limited. For example, after capturing an image, as illustrated in FIG. 6, of a cross-section of the multilayer electronic component **100** in the first and second directions from a center in the third direction with a scanning electron microscope (SEM) at a magnification of 20,000, a line profile may be performed in the image along a line **L1** from a portion of the first base electrode layer to the Ni plating layer using SEM-EDS, to obtain a graph for element amounts as illustrated in FIG. 7, and the graph may be analyzed to measure an element amount of the first alloy layer and an element amount of the second alloy layer.

(76) Referring to FIG. 6, it is possible to determine whether or not the alloy layers **131b1** and **131b2** are formed by SEM images, and referring to FIG. 7, the first alloy layer **131b1** and the second alloy layer **131b2** may be distinguished based on a point in which an Sn amount is highest in the alloy layer.

(77) A method of forming the alloy layer (**131b** and **132b**) is not particularly limited. For example, after forming a thin Sn layer on the base electrode layer (**131a** and **132a**), an Ni plating layer may be formed on the Sn layer, and may be heat-treated, as a first element of the base electrode layer (**131a** and **132a**) and an Sn layer of the Ni plating layer are diffused, the alloy layer (**131b** and **132b**) may be formed.

(78) In an embodiment, Sn included in the alloy layer (**131b** and **132b**) may be provided as an alloy with at least one of the first metal element or Ni. For example, a layer formed of Sn may not be disposed in the alloy layer (**131b** and **132b**), and Sn may be provided only as an alloy. When a layer formed of Sn is disposed in the alloy layer (**131b** and **132b**), since the layer formed of Sn has a low hydrogen permeability, there may be a risk of deterioration of insulation resistance. When a layer formed of Sn is disposed in the alloy layer (**131b** and **132b**), it may be difficult to remove residual hydrogen from the body due to the low hydrogen permeability of Sn. Therefore, in the alloy layer, Sn may be provided only as an alloy, and may be provided as an alloy with at least one of the first metal element or Ni.

(79) A sample chip for an inventive example in which a layer formed of Sn was not provided in an alloy layer (**131b** and **132b**) and a sample chip for a comparative example in which a layer formed of Sn remains in an alloy layer were prepared, and then cross-sections of each of the sample chips were photographed with a scanning electron microscope. FIG. 8 is an image of a cross-section of

an external electrode of an inventive example, scanned with a scanning electron microscope, and FIG. 9 is an image of a cross-section of an external electrode of a comparative example, scanned with a scanning electron microscope. Referring to FIG. 9, in a comparative example, it can be seen that a layer formed of Sn remains between an Ni plating layer and a base electrode layer, and a compositional difference on an interface between the Ni plating layer, the base electrode layer, and the layer formed of Sn may be clearly distinguished. Referring to FIG. 8, in an inventive example, it can be seen that a layer formed of Sn is not provided between an Ni plating layer and a base electrode layer. In an alloy layer, it can be confirmed by performing a line profile using SEM-EDS as illustrated in FIG. 7, after scanning at a high magnification as illustrated in FIG. 6.

(80) After preparing forty (40) sample chips each of the inventive example and the comparative example, a voltage of 15V was applied for 20 hours at a temperature of 85° C. and a humidity of 85%, then insulation resistance was measured, and the number of sample chips corresponding to be less than $10^7\Omega$, 10^7 to $10^8\Omega$, and more than $10^8\Omega$ was described.

(81) TABLE-US-00001 TABLE 1 Example less than $10^7\Omega$ $10^7\Omega$ to $10^8\Omega$ more than $10^8\Omega$ Inventive 1 4 35 Comparative 6 21 13

(82) It can be seen that the number of chips of which insulation resistance was measured to be less than $10^7\Omega$ was only one (1) in the inventive example and six (6) in the comparative example, and a significant difference occurred in terms of moisture resistance reliability.

(83) In addition, it can be seen that the number of chips measured with insulation resistance of $10^7\Omega$ to $10^8\Omega$ was twenty one (21) and IR Drop occurred in the majority of the sample chips in the comparative example, the number of chips measured with insulation resistance of $10^7\Omega$ to $10^8\Omega$ was four (4) and IR Drop did not occur in the majority of the sample chips in the inventive example, and a significant difference occurred in terms of moisture resistance reliability.

(84) In addition, as a result of measuring a hydrogen amount in each of the sample chips of the inventive example and a hydrogen amount in each of the sample chips of the comparative example using a hydrogen analyzer, it was measured to be 7.8 wt ppm in the inventive example and 18.4 wt ppm in the comparative example. When a layer formed of Sn was disposed as in the comparative example, it can be confirmed that it is difficult to remove residual hydrogen in the body due to low hydrogen permeability of Sn.

(85) An average thickness of an alloy layer (**131b** and **132b**) does not need to be particularly limited. When the alloy layer (**131b** and **132b**) is formed by heat-treating an Sn layer disposed between a base electrode layer (**131a** and **132a**) and an Ni plating layer (e.g., **131c1**), and the average thickness of the alloy layer (**131b** and **132b**) exceeds 0.2 μm , there may be a possibility that the Sn layer remains in the alloy layer (**131b** and **132b**). Therefore, since insulation resistance may be reduced, the average thickness of the alloy layer (**131b** and **132b**) may be 0.2 μm or less. A lower limit of the average thickness of the alloy layer (**131b** and **132b**) need not be particularly limited, but when the average thickness is less than 0.01 μm , effect of securing adhesion may be insufficient. Therefore, the average thickness may be 0.01 μm or more.

(86) In addition, referring to FIG. 5, if the average thickness of the first alloy layer (**131b1**) is referred to as tb1 and the average thickness of the second alloy layer (**131b2**) is referred to as tb2, tb1 and tb2 may be 0.1 μm or less, respectively. The average thickness of the alloy layer (**131b** and **132b**) may be the sum of tb1 and tb2.

(87) tb1 and tb2 may be measured by capturing an image of a cross-section of the multilayer electronic component **100** in the first and second directions from a center in the third direction with a scanning electron microscope (SEM) at a magnification of 20,000. More specifically, an average value may be measured by measuring a thickness of the first alloy layer and a thickness of the

second alloy layer at a plurality of points of the alloy layer (**131b** and **132b**), for example, at ten (10) points equally spaced in the first direction. The ten (10) equally spaced points may be designated in a central portion of the external electrode in the first direction.

(88) The base electrode layer (**131a** and **132a**) may be formed using any material as long as they have electrical conductivity, such as metal or the like, and a specific material may be determined in consideration of electrical characteristics, structural stability, or the like, and may further have a multilayer structure.

(89) For example, the base electrode layer (**131a** and **132a**) may be a firing electrode containing a first metal element and glass, or may be a resin-based electrode containing the first metal element and a resin.

(90) In addition, the base electrode layer (**131a** and **132a**) may have a configuration in which the firing electrode and the resin-based electrode are sequentially formed on a body. In addition, the base electrode layer (**131a** and **132a**) may be formed by transferring a sheet containing a conductive metal onto the body or by transferring a sheet containing a conductive metal onto the firing electrode.

(91) A material having excellent electrical conductivity may be used as the first metal element included in the base electrode layer (**131a** and **132a**), and is not particularly limited. Since it is difficult to implement an alloy layer according to an embodiment of the present disclosure when the first metal element is Ni and Sn, the first metal element may be at least one metal element, other than Ni and Sn. As a preferred example, the first metal element may be at least one of Cu or Ag. Since an alloy with Sn is easily formed when the first metal element is at least one of Cu or Ag, it may be easy to secure the alloy layer according to the present disclosure.

(92) The plating layer (**131b** and **132b**) may include an additional plating layer (e.g., **131c2**) disposed on the Ni plating layer (e.g., **131c1**) to improve mounting characteristics.

(93) A type of the additional plating layer (**131c2**) is not particularly limited, and may be a plating layer containing at least one of Sn, Pd, or an alloy thereof, and may be formed as a plurality of layers. When the multilayer electronic component **100** is mounted on a substrate using a solder, the additional plating layer (**131c2**) may be an Sn plating layer. When the multilayer electronic component **100** is mounted on a substrate using a conductive adhesive, the additional plating layer (**131c2**) may be a Pd plating layer.

(94) The present disclosure is not limited by the above-described embodiments and accompanying drawings, but is intended to be limited by the appended claims. Therefore, various forms of substitution, modification, and change will be possible by those skilled in the art within the scope of the technical spirit of the present disclosure described in the claims, and this will also be said to fall within the scope of the present disclosure.

(95) In addition, the expression ‘an embodiment’ does not indicate the same embodiment, and is provided to emphasize and describe different unique characteristics. an embodiment presented above is not excluded from being implemented in combination with features of another embodiment. For example, even if a matter described in one detailed embodiment is not described in another embodiment, and it can be understood as a description related to another embodiment, unless there is a description contradicting or contradicting the matter in another embodiment.

(96) Terms used in this disclosure may be only used to describe an embodiment, and may not be intended to limit the disclosure. In this case, singular expressions include plural expressions unless the context clearly indicates otherwise.

(97) As one effect of various effects of the present disclosure, an alloy layer may be disposed between a base electrode layer and an Ni plating layer, to improve adhesion between the base electrode layer and the Ni plating layer.

(98) Various advantages and effects of the present disclosure are not limited to the above description, and will be more easily understood in the process of describing specific embodiments of the present disclosure.

(99) While example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A multilayer electronic component comprising: a body including a dielectric layer and an internal electrode; and an external electrode including a base electrode layer disposed on the body and including a first metal element, and an alloy layer and a plating layer, sequentially disposed on the base electrode layer, wherein the alloy layer includes a first alloy layer disposed to contact the base electrode layer and including an alloy of the first metal element and Sn, and a second alloy layer disposed to contact the first alloy layer and including an alloy of Ni and Sn, and the plating layer includes an Ni plating layer disposed to contact the second alloy layer.
2. The multilayer electronic component of claim 1, wherein the alloy layer has the highest Sn amount on an interface between the first alloy layer and the second alloy layer.
3. The multilayer electronic component of claim 1, wherein a first metal element amount of the first alloy layer is highest on an interface with the base electrode layer, and is lowest on an interface with the second alloy layer.
4. The multilayer electronic component of claim 1, wherein a first metal element amount of the first alloy layer decreases from an interface with the base electrode layer toward an interface with the second alloy layer.
5. The multilayer electronic component of claim 1, wherein a Ni amount of the second alloy layer is highest on an interface with the Ni plating layer, and is lowest on an interface with the first alloy layer.
6. The multilayer electronic component of claim 1, wherein a Ni amount of the second alloy layer decreases from an interface with the Ni plating layer toward an interface with the first alloy layer.
7. The multilayer electronic component of claim 1, wherein a first metal element amount of the alloy layer is highest on an interface with the base electrode layer, and is lowest on an interface with the Ni plating layer, and a Ni amount of the alloy layer is lowest on an interface with the base electrode layer, and is highest on an interface with the Ni plating layer.
8. The multilayer electronic component of claim 1, wherein a first metal element amount of the alloy layer decreases from an interface with the base electrode layer toward an interface with the Ni plating layer, and a Ni amount of the alloy layer increases from an interface with the base electrode layer toward an interface with the Ni plating layer.
9. The multilayer electronic component of claim 8, wherein a Sn amount of the alloy layer increases from an interface with the base electrode layer toward an interface between the first alloy layer and the second alloy layer, and decreases from an interface between the first alloy layer and the second alloy layer toward an interface with the Ni plating layer.
10. The multilayer electronic component of claim 1, wherein Sn included in the alloy layer is provided as an alloy with at least one of the first metal element or Ni.
11. The multilayer electronic component of claim 1, wherein an average thickness of the alloy layer is 0.2 μm or less.
12. The multilayer electronic component of claim 1, wherein an average thickness of the first alloy layer and an average thickness of the second alloy layer are 0.1 μm or less, respectively.
13. The multilayer electronic component of claim 1, wherein the first metal element is at least one of metal elements other than Ni and Sn.
14. The multilayer electronic component of claim 1, wherein the first metal element is at least one of Cu or Ag.
15. The multilayer electronic component of claim 1, wherein the plating layer further comprises a Sn plating layer disposed on the Ni plating layer.

16. A multilayer electronic component comprising: a body including a dielectric layer and an internal electrode; and an external electrode including a base electrode layer disposed on the body and including a first metal element, and an alloy layer and a Ni plating layer, sequentially disposed on the base electrode layer, wherein the alloy layer includes an alloy of the first metal element, Sn, and Ti, and an average thickness of the alloy layer is 0.2 μm or less.
17. The multilayer electronic component of claim 16, the average thickness of the alloy layer is 0.01 μm or more.
18. The multilayer electronic component of claim 16, wherein the alloy layer has the highest Sn amount in a region away from the Ni plating layer and the base electrode layer.
19. The multilayer electronic component of claim 16, wherein the alloy layer has the highest first metal element amount on an interface with the base electrode layer and has the lowest first metal element amount on an interface with the nickel layer.
20. The multilayer electronic component of claim 16, wherein a first metal element amount in the alloy layer decreases from an interface with the base electrode layer toward an interface with the Ni plating layer.
21. The multilayer electronic component of claim 16, wherein the alloy layer has the highest Ni amount on an interface with the Ni plating layer and has the lowest Ni amount on an interface with the base electrode layer.
22. The multilayer electronic component of claim 16, wherein a Ni amount in the alloy layer increases from an interface with the base electrode layer toward an interface with the Ni plating layer.
23. The multilayer electronic component of claim 16, wherein the first metal element includes at least one of Cu or Ag.
24. The multilayer electronic component of claim 16, wherein the alloy layer is in contact with the Ni plating layer and the base electrode layer.
25. The multilayer electronic component of claim 16, wherein the external electrode layer further comprises a Sn plating layer disposed on the Ni plating layer.
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